

CE 564: Fundamentals of Fracture Mechanics

Course Code: CE 564

Course Name: Fundamentals of Fracture Mechanics

L-T-P-C: 2.5-0.5-0-3

Prerequisite: IC 242 Continuum Mechanics, IC 240 Mechanics of Rigid Bodies, IC 241 Materials Science for Engineers

Students intended for: UG/PG

Elective or Core: Elective

Mutual Exclusion: ME 506

Approval: 8th Senate (ME 506), 59th BoA (CE 564)

Course contents

- **Introduction:** Why structures fail?, An atomic view, Energy criterion, Stress intensity, Effect of material properties on fracture, Modes of failure. [4 Lectures]
- **Linear Elastic Fracture Mechanics :** An atomic view of fracture, Effect of flaws on stress concentration, Griffith theory of fracture, Energy release rate, Instability and the R curve, Stress analysis of cracks, Stress Intensity Factor (SIF) , Determination SIF of different geometries, Crack tip plasticity, Irwin approach, Plane strain fracture, Mixed mode fracture. [10 Lectures]
- **Elastic-Plastic Fracture Mechanics:** Crack tip opening displacement (CTOD), J contour integral, Relationship between J and CTOD, Resistance curve, Cleavage fracture, failure criterion, Three-dimensional effect, Crack arrest. [10 Lectures]
- **Fracture Mechanisms in Metals and Nonmetals :** Ductile fracture, Void nucleation, Void growth, coalescence, Ductile crack growth, Brittle Fracture, Cleavage, Mechanisms of cleavage initiation, Transgranular and Intergranular fracture, Fracture Mechanisms in Nonmetals. [8 Lectures]
- **Fracture Toughness Testing:** Specimen configurations and orientations, KIC testing, CTOD testing, Measurement of J-critical, Determination of critical G in Mode I and Mode II. [4 Lectures]
- **Fatigue Crack Propagation and Environmental Assisted Fracture Failure:** Fatigue crack growth, Crack closure, A short introduction to environmental assisted fracture failure. [4 Lectures]
- **Computational Fracture Mechanics:** Modeling and analysis [2 Lectures]

Text Books:

1. T.L. Anderson, **Fracture Mechanics: A Fundamentals and Applications**, 3rd Edition, CRC Press, 2005.

References:

1. Prashant Kumar, **Elements of Fracture Mechanics**, Tata McGraw Hill, New Delhi, India.
2. G.E. Dieter, **Mechanical Metallurgy**, McGraw Hill, 2009